Oil content

In asphalt quality work, “oil content” commonly refers to asphalt binder content. It does not mean engine oil. Binder content affects the mixture, but its reported percentage is meaningless unless the mass basis and test method are known.

> Define Section 502

LEARNING STEP 1

Define Section 502

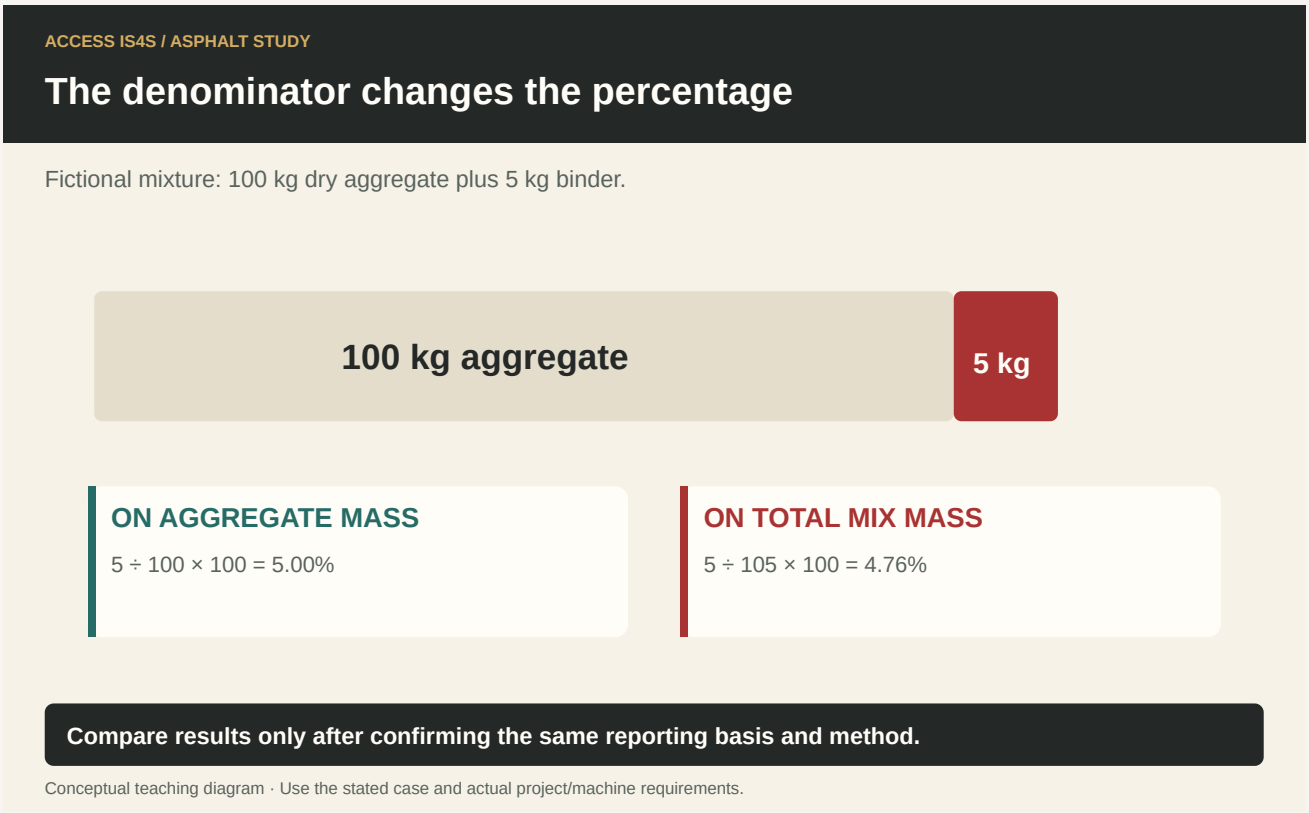
Section 502.12 of the supplied 2006 specification evaluates actual asphalt content against the approved content. Table 502-F expresses deviation in kilograms of binder per 100 kg of dry aggregate. This is a different basis from binder as a percentage of total mixture. For example, 5 kg binder with 100 kg dry aggregate is 5.00% on aggregate mass and 5/105 × 100 = 4.76% on total mixture mass. Never subtract results reported on different bases.

The historical procedure uses the sample mean from the specified sub-lot samples. Preserve sample identity, test method, lot and lift. Compute actual minus approved content, then use the magnitude of the deviation where the table applies to either above or below. Follow the contract’s rounding and reporting rules rather than rounding early to reach a favourable band.

In Table 502-F, a deviation of 0–0.15 earns +$0.50/t, 0.16–0.30 has no adjustment, and 0.31–0.35 gives –$1/t. Larger deviations have larger deductions and different top/lower-lift rejection boundaries. The full table and associated clauses govern; these are historical exercise values, not current project permission. A top-lift deviation greater than 0.50 triggers the stated rejection/remedial provisions. An operator does not fix a field result by independently changing the approved JMF.

Possible causes of an unexpected result include actual production variation, sampling problems, incorrect identities, test issues or a mistaken reporting basis. Preserve the original evidence and refer the issue to the quality lead. A single favourable retest is not automatic authority to discard an unfavourable result.

The denominator changes the percentage



Note. ACCESS IS4S. (2026). The denominator changes the percentage [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

2006 Table 502-F: binder-content adjustment

TABLE 502-F – PAYMENT ADJUSTMENTS FOR ASPHALT CONTENT

Deviation of the Actual Asphalt Content from the Approved Asphalt Content (either above or below) kg/100kg of Dry Aggregate	Payment Adjustment for Asphalt Content $ per tonne	
	Top Lift	Lower Lift
From 0 to 0.15	$0.50	$0.50
From 0.16 to 0.30	$0.00	$0.00
From 0.31 to 0.35	-$1.00	-$1.00
From 0.36 to 0.40	-$2.00	-$2.00
From 0.41 to 0.45	-$3.00	-$3.00
From 0.46 to 0.50	-$4.00	-$4.00
From 0.51 to 0.55	Reject	-$5.00
0.56 and greater	Reject	Reject
For top lift deviations of more than 0.50 the Contractor shall either overlay or remove and replace the Lot. For lower lift deviations of more than 0.55, no payment will be made and the Ministry Representative will determine whether removal and replacement is necessary.		

Note. British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502 (section p. 12; PDF p. 308). Historical contract-reading example; complete clauses and actual project adoption govern. Full source context.

Worked example · Read the basis before the band

Approved binder content is 5.20 kg/100 kg dry aggregate. The three fictional sub-lot results are 5.45, 5.50 and 5.49 on that same basis. Lot mass is 500 t.

Think it through.

Mean = (5.45 + 5.50 + 5.49)/3 = 5.48. Absolute deviation = 0.28. Under the dated Table 502-F exercise, 0.16–0.30 gives $0/t, so this isolated adjustment is $0. This says nothing about other acceptance parameters.

Try it and explain your thinking

Approved content is 5.20 and the mean is 4.86, both kg/100 kg dry aggregate

Approved content is 5.20 and the mean is 4.86, both kg/100 kg dry aggregate. Find deviation and the historical isolated adjustment for 300 t.

Skills you use: Numeracy, Reading.

A report says 5.2% but has no basis

A report says 5.2% but has no basis. Can it be compared directly with 5.2 kg/100 kg dry aggregate?

Skills you use: Problem solving, Communication.

Write a two-line hold note for conflicting sample IDs.

Write a two-line hold note for conflicting sample IDs.

Skills you use: Writing, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify binder basis and lot/lift before calculation.	0 / 1 / 2	Evidence/help: _____
Read the complete applicable payment/rejection table.	0 / 1 / 2	Evidence/help: _____
Escalate a conflicting result without altering original evidence.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G2, printed p. 53; source PDF pp. 55–55.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 2: Petroleum Asphalt; source PDF pp. 55–142. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 4: Hot Mix Asphalt (HMA) Design; source PDF pp. 193–280. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

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Study chapter · Outline G3

Application

Application rate describes how much asphalt mixture was placed over a measured area. It is a mass-per-area quantity. It can help identify a quantity or thickness concern, but does not replace density, level or compacted-thickness checks.

Define Section 502

LEARNING STEP 1

Define Section 502

Section 502.14 of the 2006 specification determines application rate from the delivered tonnage assigned to the lot and its covered area, with the specified treatment of entrance letdowns and intersections. Reconcile weigh tickets, returned material, location and lift according to the contract. Counting an entire truck twice or including a ticket from another lane changes the result even when the arithmetic is correct.

Use consistent units: 1 tonne = 1,000 kg. Actual application rate in kg/m² = accepted assigned mass in kg ÷ measured area in m². The ratio used by Table 502-H is actual rate ÷ specified rate × 100. This ratio is a percentage of the specified rate, not a binder percentage and not percent density.

The dated table distinguishes bottom/single lifts from the top lift of multiple lifts. For the uncomplicated 96.0–103.9% band, both columns show +$0.50/t. Other bands include different deductions, restrictions on payment for excess material and rejection/remedial provisions. Read the full row and correct lift column. Do not treat over-application as a desirable way to earn payment: it can change grade, tie-ins and material demand.

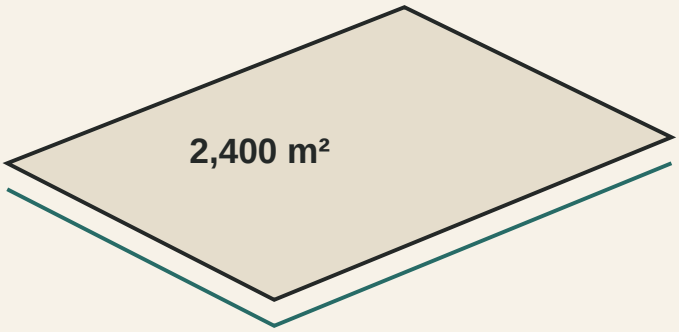
If a calculated rate seems wrong, check area boundaries, units, tickets, lift identity and authorized changes first. Then compare with thickness and survey evidence. A rate alone cannot prove a uniform mat; a thick patch and a thin patch can average to the specified rate.

Mass divided by area gives application rate

ACCESS IS4S / ASPHALT STUDY

Mass divided by area gives application rate

Fictional lot: 288 t over 2,400 m²; specified rate 120 kg/m².



CALCULATE

288 t = 288,000 kg
288,000 ÷ 2,400
= 120 kg/m²

120 ÷ 120 × 100 = 100% of the specified rate

An average rate does not prove uniform thickness or full acceptance.

Conceptual teaching diagram · Use the stated case and actual project/machine requirements.

Note. ACCESS IS4S. (2026). Mass divided by area gives application rate [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

2006 Table 502-H: application-rate adjustment

TABLE 502-H – PAYMENT ADJUSTMENTS FOR MATERIAL APPLICATION RATE

Actual Application Rate Expressed as % of Specified Application Rate	Assessment for Application Rate $ per tonne	
	Bottom Lift or Single Lift	Top Lift of Multiple Lifts
110 +	-$6.00 for all material in the Lot up to 110% and no payment for product in excess of 110.0%	No payment
106.0 – 109.9	-$4.00 for all material in the Lot	-$6.00 for all material in the Lot up to 106% and no payment for product in excess of 106.0%
105.0 – 105.9		-$4.00 for all material in the Lot
104.0 – 104.9	$0.00	-$2.00 for all material
96.0– 103.9	+$0.50	+ $0.50
94.0 - 95.9	-$1.00	-$1.00
92.0 - 93.9	-$2.00	-$2.00
90.0 - 91.9	-$3.00	-$3.00
85.0 – 89.9	-$5.00	-$5.00
84.9 or lower	Rejected, Mill and Fill, and/or rejected with no remedial work required at the discretion of the Ministry Representative	Rejected, Mill and Fill, Overlay, and/or rejected with no remedial work required at the discretion of the Ministry Representative

Note. British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502 (section p. 14; PDF p. 310). Historical contract-reading example; complete clauses and actual project adoption govern. Full source context.

Worked example · Mass over area

A fictional lot receives 288 t over 2,400 m². Specified application is 120 kg/m².

Think it through.

288 t = 288,000 kg. Actual = 288,000/2,400 = 120 kg/m². Ratio = 120/120 × 100 = 100%. In the dated 96.0–103.9% band, isolated adjustment = 288 × $0.50 = +$144, subject to all applicable conditions.

Try it and explain your thinking

A 200 m × 3.5 m rectangle receives 77 t

A 200 m × 3.5 m rectangle receives 77 t. Find actual application rate.

Skills you use: Numeracy, Reading.

Specified rate is 115 kg/m²

Specified rate is 115 kg/m². What percentage of specified is 110 kg/m²?

Skills you use: Numeracy, Problem solving.

Design a ticket register that a partner can audit.

Design a ticket register that a partner can audit.

Skills you use: Digital, Writing, Collaboration, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Measure and document a lot boundary.	0 / 1 / 2	Evidence/help: _____
Reconcile unique tickets to the correct lift.	0 / 1 / 2	Evidence/help: _____
Calculate rate and explain why average rate does not prove uniform thickness.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G3, printed p. 54; source PDF pp. 56–56.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

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Study chapter · Outline G4

Segregation

Segregation is separation within the mixture that produces an uneven distribution of aggregate sizes or other mixture characteristics. A visibly coarse patch can be a clue, but a crew needs agreed inspection criteria and a traceable location before deciding its cause or treatment.

Define Section 502

LEARNING STEP 1

Define Section 502

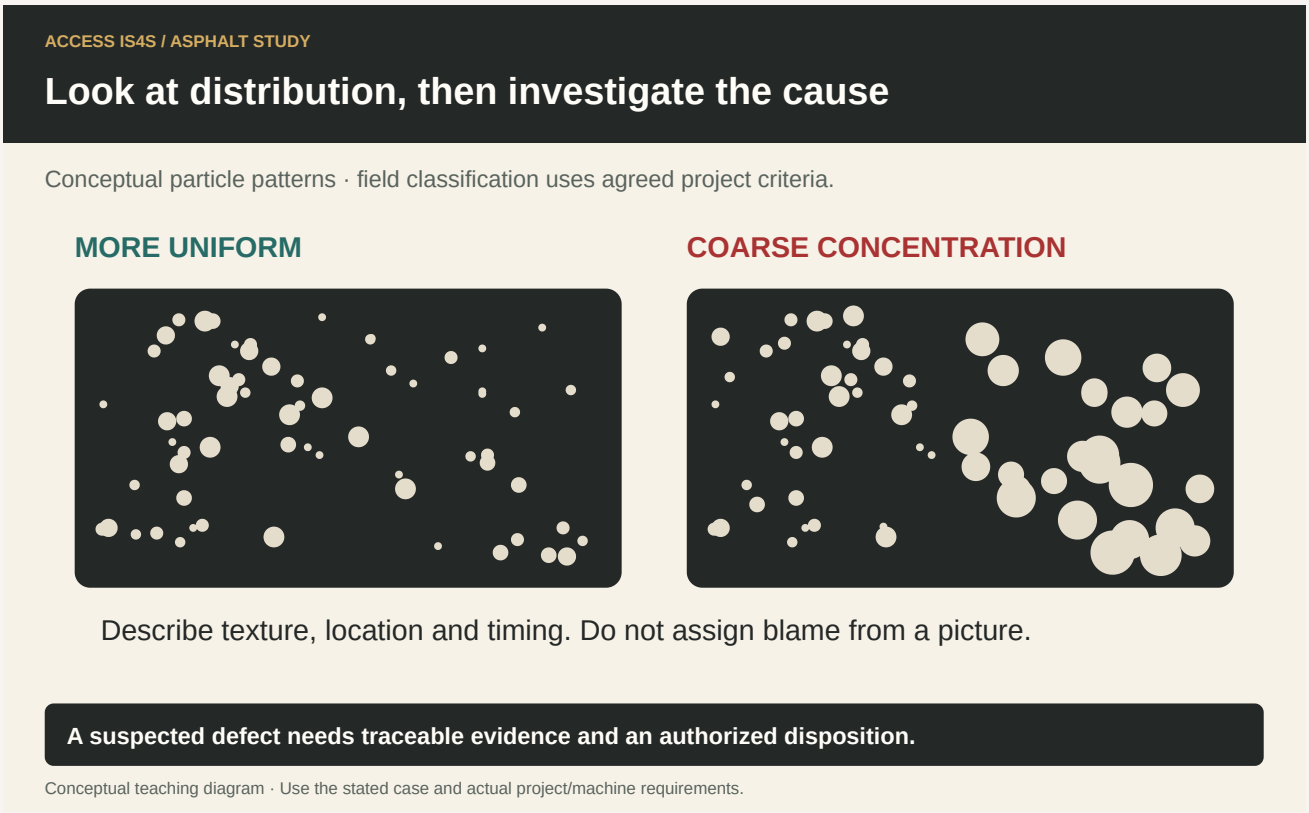
Section 502.15 of the 2006 specification calls for a homogeneous mixture and uniform surface texture. Its visual classification distinguishes slight, moderate and severe segregation using the amount of stone and remaining binder/fine-aggregate matrix. The project's agreed comparison areas or photographs support consistent judgement. Temperature differences can also affect paving behaviour; thermal segregation and aggregate-size segregation are related quality concerns but are not interchangeable descriptions.

Track where a pattern occurs: at truck changes, the end of loads, edges, behind a particular part of the screed or at repeated stops. Material handling, stockpiles, truck loading/unloading, hopper management and the feed system all deserve investigation. Do not claim a single cause from a photograph alone. Record station, lane, lift, time, load and observed texture, then notify the responsible lead while the crew can still investigate the process.

The historical top-lift tables 502-I-1 and I-2 address remediation and payment. They include different remedies by severity and counts per lane kilometre. These dated repair descriptions are an exercise in reading a contract, not a field instruction to apply slurry or overlay without approval. Current project specifications and the authorized repair design govern actual work. A payment deduction does not automatically remove the requirement for remediation.

Prevention is a coordinated task. Maintain the specified material-handling and paving process, communicate supply changes early, and avoid scattering coarse material onto a finished surface. When a defect is suspected, preserve evidence and obtain direction before an improvised repair hides the cause.

Look at distribution, then investigate the cause



Note. ACCESS IS4S. (2026). Look at distribution, then investigate the cause [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Segregation along the outside edge of a mat



Figure 8.26 Segregation Along Outside Edge of Mat

Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4, Fig. 8.26, p. 402; PDF p. 436). Faithful scan crop; original figure caption retained. Original image contributor: James A. Scherocman, P.E., Consulting Engineer; credit register, PDF pp. 803–805. Full source context.

Worked example · Describe before diagnosing

A coarse-looking strip appears near the right edge after three truck changes. The surrounding surface has more fine matrix.

Think it through.

Record the repeated location and timing, photograph from a safe position with a scale and location label, and notify the paving/quality lead. Compare with agreed criteria and inspect the process under authorized conditions. Do not declare severe segregation or choose a repair from appearance alone.

Try it and explain your thinking

Write three useful observations and one unsupported conclusion for the case.

Write three useful observations and one unsupported conclusion for the case.

Skills you use: Writing, Problem solving.

In the dated Table 502-I-2, a 1 lane-km top lift has 4 slight areas, zero moderate and zero severe

In the dated Table 502-I-2, a 1 lane-km top lift has 4 slight areas, zero moderate and zero severe. What is the isolated adjustment?

Skills you use: Numeracy, Reading.

Propose a safe digital defect log and a respectful crew discussion.

Propose a safe digital defect log and a respectful crew discussion.

Skills you use: Digital, Communication, Collaboration, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Compare texture using approved project criteria.	0 / 1 / 2	Evidence/help: _____
Locate and record a suspected segregated area.	0 / 1 / 2	Evidence/help: _____
Separate observation, possible cause and authorized disposition.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G4, printed p. 55; source PDF pp. 57–57.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

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Density

Percent density compares measured in-place density with a defined reference density. The reference is part of the result. Saying only “the density is 94%” leaves out information needed to interpret it.

Define Section 502

LEARNING STEP 1

Define Section 502

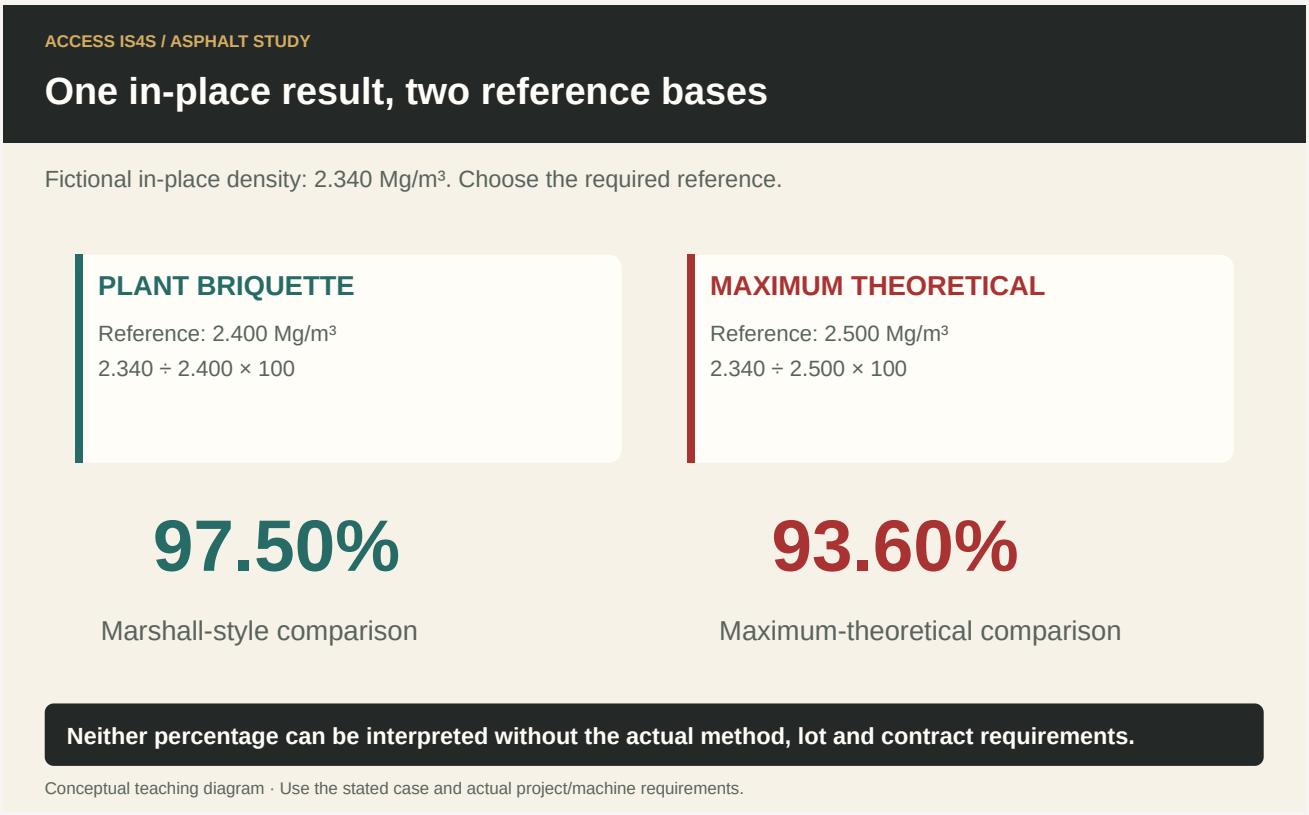
Section 502.11 of the supplied 2006 document uses different references for Marshall and Superpave mixtures. The Marshall expression divides in-place density by the density of the plant briquette. The Superpave expression divides in-place density by maximum theoretical density determined by the specified procedure. Use compatible units, the applicable method and representative results from the correct lot.

A core and a nuclear gauge do not produce evidence through the same procedure. Follow the approved sampling, testing, calibration/correlation and reporting requirements. Nuclear gauge operation requires the applicable CNSC licensing, training and controls; reading this chapter does not authorize it. Record location, lift, sample/gauge identification, method, density units and reference result. Avoid substituting a different day’s reference because it produces a more favourable percentage.

The dated tables E-1 and E-2 use different acceptance/payment bands. For example, a Marshall lot average of 97.6–98.0% has a +$0.50/t adjustment, while the Superpave +$0.50/t band is 93.0–93.9%. These bands cannot be exchanged. Sub-lot limits, rejection provisions and the other acceptance requirements also apply. A satisfactory lot mean can conceal an unacceptable individual area.

Compaction effort, temperature, mixture behaviour, support and rolling pattern affect the outcome. More passes are not automatically better: they may be ineffective after cooling or damage an unstable mat. Use the approved pattern and quality feedback to guide authorized changes.

One in-place result, two reference bases



Note. ACCESS IS4S. (2026). One in-place result, two reference bases [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

2006 Tables 502-E-1 and E-2: different density bases

TABLE 502-E-1 – PAYMENT ADJUSTMENTS FOR DENSITY – MARSHALL

% Density Lot Average	Payment Adjustment ($ Per Tonne) for Design Lift Thickness
98.6 Plus	+$1.00
98.1 – 98.5	+$0.75
97.6 – 98.0	+$0.50
97.0 – 97.5	+$0.25
96.6 – 96.9	-$0.50
96.0 – 96.5	-$1.00
Less than 96.0%	Reject

If any Sub-Lot has a density below 96%, the Sub-Lot shall be considered rejectable, and, subject to the Contractor's opportunity to increase the density as described in DBSS 502.11.04, shall be rejected.

TABLE 502-E-2 – PAYMENT ADJUSTMENTS FOR DENSITY - SUPERPAVE

% Density Lot Average	Payment Adjustment ($ Per Tonne) for Design Lift Thickness
95.0 Plus	+$1.00
94.0 – 94.9	+$0.75
93.0 – 93.9	+$0.50
92.0 – 92.9	+$0.25
91.6 – 91.9	-$0.50
91.0 – 91.5	-$1.00
less than 91.0 %	Reject

If any Sub-Lot has a density below 91%, the Sub-Lot shall be considered rejectable, and, subject to the Contractor's opportunity to increase the density as described in DBSS 502.11.04, shall be rejected. Percent Density is based on a comparison to the Maximum Theoretical Density.

Note. British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502 (section p. 11; PDF p. 307). Historical contract-reading example; complete clauses and actual project adoption govern. Full source context.

Worked example · Two legitimate references, two different percentages

In-place density is 2.340 Mg/m³. A fictional plant briquette result is 2.400 Mg/m³; maximum theoretical density is 2.500 Mg/m³.

Think it through.

Marshall-style comparison: 2.340/2.400 × 100 = 97.50%. Maximum-theoretical comparison: 2.340/2.500 × 100 = 93.60%. Both calculations are arithmetic examples; use only the reference required for the actual mixture and contract.

Try it and explain your thinking

For 2.325 Mg/m³ in place and 2.500 Mg/m³ maximum theoretical, calculate percent.

For 2.325 Mg/m³ in place and 2.500 Mg/m³ maximum theoretical, calculate percent.

Skills you use: Numeracy, Reading.

A 400 t fictional Superpave lot average is 93.4%

A 400 t fictional Superpave lot average is 93.4%. Find the isolated dated table adjustment, then name two checks still needed.

Skills you use: Numeracy, Problem solving.

A spreadsheet divides every test by a fixed 2.400

A spreadsheet divides every test by a fixed 2.400. What should a reviewer do?

Skills you use: Digital, Collaboration, Writing.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Explain and select the correct density denominator.	0 / 1 / 2	Evidence/help: _____
Trace test results to location and reference.	0 / 1 / 2	Evidence/help: _____
Distinguish a lot average from individual acceptance requirements.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G5, printed p. 56; source PDF pp. 58–58.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 9: Hot Mix Asphalt Compaction; source PDF pp. 447–478. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- Troxler Electronic Laboratories. (Edition 1.3, February 2021). RoadReader Model 3430 Surface Moisture-Density Gauge: Manual of Operation and Instruction. Open publication.
- Canadian Nuclear Safety Commission. (Web snapshot: October 30, 2026). Portable Nuclear Gauges. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline G6

Smoothness

Smoothness concerns the longitudinal profile experienced by a moving vehicle. It differs from crossfall, compacted thickness and density. A mat can meet one of those requirements while failing another.

Define Section 502

LEARNING STEP 1

Define Section 502

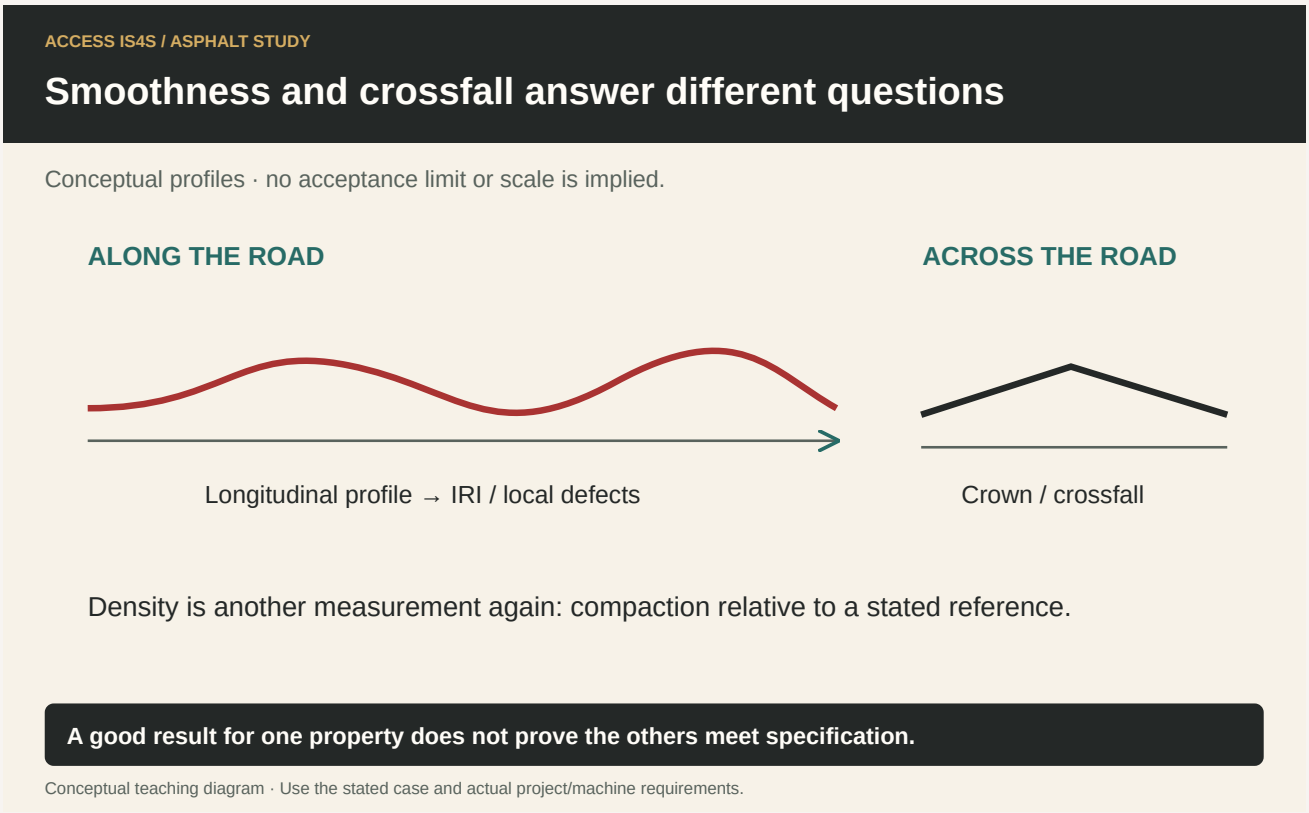
Section 502.16 of the dated 2006 specification measures longitudinal profile with a specified precision rolling instrument and computes the International Roughness Index (IRI). IRI is commonly expressed in m/km; lower values indicate a smoother profile under the model. It is not a direct measurement of crossfall or a single bump's height. Straightedge checks and localized-defect requirements answer different questions and may apply in addition to IRI.

Identify the lane, direction, station limits, lift and applicable lot/sub-lot rules. The historical clauses treat auxiliary lanes and short lengths specifically, and use the applicable Table 502-J or J-2 when adopted. Acceptance limits and payment schedules depend on the road classification and contract. Do not import an IRI threshold from another job. Preserve the instrument record and the quality lead's interpretation.

Smoothness begins before the profiler arrives: prepared base, correct setup, steady paving speed and material head, coordinated truck changes, careful joints and suitable rolling practice all matter. Abrupt control changes, stops and poorly matched joints can leave profile defects. A roller cannot reliably erase every paver setup error. Report an emerging pattern early enough for the lead to investigate the process.

The 2006 auxiliary-lane example prorates a qualifying adjustment by segment length relative to a 1 km lot. This is a contract rule, not a general conversion from IRI to dollars. Repairs and remeasurement require authorized procedures. Keep the original and subsequent profiles linked so the correction can be evaluated transparently.

Smoothness and crossfall answer different questions



Note. ACCESS IS4S. (2026). Smoothness and crossfall answer different questions [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

2006 Tables 502-J and J-2: smoothness

TABLE 502-J – LOT ASSESSMENT AND PAYMENT ADJUSTMENTS FOR SMOOTHNESS

For the final surface course only, the following payment adjustments shall apply to each Lot.

Lot IRI (m/km)	Payment Adjustment
≤ 0.80	+$2,000
0.81 to 0.90	+$1000
0.91 to 1.00	+$500
1.01 to 1.10	+$200
1.11 to 1.20	0
1.21 to 1.30	-$100
1.31 to 1.40	-$250
1.41 to 1.50	-$600
1.51 to 1.60	-$1,400
1.61 to 1.70	-$2,000
1.71 to 1.80	-$3,000
1.80	**ACCEPTANCE LIMIT.** Reject or corrective work required above this level

The Lot IRI is the average IRI value calculated for the Sub-Lots within the Lot.

TABLE 502-J-2 – LOT ASSESSMENT AND ALTERNATIVE PAYMENT ADJUSTMENTS FOR SMOOTHNESS

For the final surface course only, and only where specifically invoked by the Contract, the following payment adjustments shall apply to each Lot on a road designated by the Ministry for application of alternative smoothness payment adjustments

Lot IRI (m/km)	Payment Adjustment
<1.00	+$2000
1.01 – 1.10	+$1000
1.11 – 1.20	+$500
1.21 – 1.30	+$250
1.31 – 1.40	0
1.41 – 1.60	-$300
1.61 – 1.80	-$750
1.81 – 1.90	-$1500
1.90 – 2.00	-$2500
2.00	**ACCEPTANCE LIMIT.** Reject or corrective work required above this level

The Lot IRI is the average IRI calculated for the Sub-Lots within the Lot

Note. British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502 (section p. 16; PDF p. 312). Historical contract-reading example; complete clauses and actual project adoption govern. Full source context.

Worked example · Prorating an authorized historical adjustment

For a fictional exercise, the applicable table and acceptance checks establish a +$800 adjustment for a full 1 km lot. The qualifying auxiliary segment is 565 m.

Think it through.

Length factor = 565/1,000 = 0.565. Prorated adjustment = 0.565 × $800 = +$452. The $800 is an explicitly supplied exercise value, not a quoted Table 502-J value or a current project entitlement.

Try it and explain your thinking

A qualifying segment is 240 m and the supplied full-kilometre adjustment is −$1,200

A qualifying segment is 240 m and the supplied full-kilometre adjustment is −$1,200. Calculate the exercise adjustment.

Skills you use: Numeracy, Reading.

A learner says a good density result proves a smooth road

A learner says a good density result proves a smooth road. Explain the gap.

Skills you use: Communication, Problem solving.

Create a handover for a repeated bump near each truck-change location.

Create a handover for a repeated bump near each truck-change location.

Skills you use: Writing, Digital, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify the profile measurement and applicable contract limits.	0 / 1 / 2	Evidence/help: _____
Separate IRI, localized defects and crossfall.	0 / 1 / 2	Evidence/help: _____
Trace a profile concern to a location and authorized corrective process.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G6, printed p. 57; source PDF pp. 59–59.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

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